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Volume 01 — Deep Learning Foundations, Autoencoders, and VAE Foundations

Lectures L01-L09 · Weeks 1-3

L01: Introduction to Generative AI and Course Overview

Learning objectives

- Distinguish generative models from discriminative models.
- State the core problem of generative modeling.
- Survey the landscape of modern generative AI.

Discriminative vs. generative modeling

A **discriminative model** learns the conditional distribution $P(y \mid x)$ —for example, classifying an image as “cat” or “dog.” A **generative model** learns the data distribution $P(x)$ or the joint distribution $P(x, y)$, enabling **sampling** of new data points.

Paradigm	Learns	Example task	Example output
Discriminative	$P(y \mid x)$	Image classification	Label
Generative	$P(x)$ or $P(x \mid c)$	Image synthesis	New image

The generative modeling problem

Given i.i.d. samples $\{x^{(1)}, \dots, x^{(N)}\}$ from an unknown distribution $p_{\text{data}}(x)$, we seek a parametric model $p_{\theta}(x)$ such that:

1. **Density estimation:** $p_{\theta}(x)$ approximates $p_{\text{data}}(x)$.

2. **Sampling:** we can draw $\tilde{x} \sim p_\theta(x)$.
3. **Likelihood:** we can evaluate $\log p_\theta(x)$ for unseen data.

Taxonomy of generative models

Explicit density	Implicit density
└─ Autoregressive	└─ GANs
└─ Flow models	└─ Score-based / Diffusion
└─ VAEs (lower bound)	

Key takeaways

- Generative AI creates new content by learning probability distributions.
 - This course progresses from tractable models (autoencoders, VAEs) to adversarial and diffusion-based approaches, then to language models.
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L02: Neural Network Fundamentals for Generative Models

Learning objectives

- Review feedforward networks, activation functions, and backpropagation.
- Understand how neural networks parameterize complex distributions.

Feedforward computation

For layer l , given input $h^{(l-1)}$:

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)})$$

where σ is a nonlinear activation (ReLU, GELU, sigmoid).

Loss and optimization

For a dataset $\mathcal{D}$, we minimize:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} \ell(f_\theta(x), y)$$

using stochastic gradient descent:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}(\theta)$$

Backpropagation

The chain rule propagates gradients backward:

$$\frac{\partial \mathcal{L}}{\partial W^{(l)}} = \frac{\partial \mathcal{L}}{\partial h^{(l)}} \cdot \frac{\partial h^{(l)}}{\partial W^{(l)}}$$

For generative models, the “target” y is often the input itself (reconstruction) or an adversarial signal.

Universal approximation

Neural networks with sufficient width/depth can approximate arbitrary continuous functions (Cybenko, 1989; Hornik et al., 1989). This justifies using deep networks as flexible density estimators and decoders.

Key takeaways

- Backpropagation is the workhorse for training all models in this course.
 - Activation choices (ReLU for hidden layers, sigmoid/tanh for bounded outputs) matter for generative quality.
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L03: Convolutional Neural Networks: LeNet, VGG, ResNet

Learning objectives

- Explain convolution, pooling, and spatial hierarchy.
- Compare LeNet, VGG, and ResNet architectures.

Convolution operation

For input X and kernel K of size $k \times k$:

$$(X * K)_{i,j} = \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} X_{i+m, j+n} \cdot K_{m,n}$$

CNNs exploit **translation equivariance** and **local connectivity**, making them ideal for image generation and encoding.

Architecture comparison

Model	Year	Key idea	Depth
LeNet-5	1998	Conv + pool + FC	5 layers

Model	Year	Key idea	Depth
VGG-16	2014	Small 3×3 filters, deep stacks	16 layers
ResNet-50	2015	Skip connections: $h^{(l+1)} = h^{(l)} + F(h^{(l)})$	50 layers

Residual connections

$$h^{(l+1)} = h^{(l)} + \mathcal{F}(h^{(l)}; W^{(l)})$$

Skip connections mitigate vanishing gradients and enable training very deep encoders/decoders—critical for VAEs and diffusion U-Nets.

Role in generative AI

CNNs serve as: - **Encoders** mapping images to latent codes (VAE, GAN discriminator input). - **Decoders** generating images from latent vectors (DCGAN generator). - **Backbones** in U-Net architectures for diffusion models.

L04: Encoder-Decoder Architecture and Bottleneck Representations

Learning objectives

- Describe the autoencoder architecture.
- Explain the role of the bottleneck latent space.

Autoencoder structure

An autoencoder consists of:

1. **Encoder** $f_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^k$ — maps input to latent code z .
2. **Decoder** $g_\theta : \mathbb{R}^k \rightarrow \mathbb{R}^d$ — reconstructs input from z .

$$\hat{x} = g_\theta(f_\phi(x))$$

Bottleneck principle

With $k \ll d$, the network must learn a **compressed representation** that captures salient features. The bottleneck forces the model to discard noise and retain structure.

Information-theoretic view

The bottleneck implements a form of **rate-distortion trade-off**: fewer bits in z mean higher compression but potentially higher reconstruction error.

Applications

- Dimensionality reduction (nonlinear PCA)
 - Pre-training representations for downstream tasks
 - Foundation for VAEs (add probabilistic latent space)
-

L05: Training Autoencoders and Reconstruction Loss

Learning objectives

- Formulate the autoencoder training objective.
- Compare loss functions for different data types.

Training objective

$$\mathcal{L}_{\text{AE}}(\theta, \phi) = \frac{1}{N} \sum_{i=1}^N \|x^{(i)} - g_{\theta}(f_{\phi}(x^{(i)}))\|^2$$

For binary images, cross-entropy is often preferred:

$$\mathcal{L}_{\text{BCE}} = - \sum_j [x_j \log \hat{x}_j + (1 - x_j) \log(1 - \hat{x}_j)]$$

Latent space geometry

After training, nearby points in latent space $\mathcal{Z}$ should correspond to semantically similar inputs. However, standard autoencoders have **holes** in $\mathcal{Z}$ —regions that do not correspond to any training example—making random sampling unreliable.

Limitations motivating VAEs

Issue	Consequence
Non-probabilistic latent space	Cannot sample meaningfully
No regularization on z	Overfitting; irregular latent geometry
Deterministic encoding	No uncertainty quantification

L06: Denoising and Sparse Autoencoders

Learning objectives

- Understand denoising autoencoders (DAE) and their regularization effect.
- Describe sparse autoencoder objectives.

Denoising autoencoder

Corrupt input $\tilde{x} = \text{corrupt}(x)$ and train to recover x :

$$\mathcal{L}_{\text{DAE}} = \|x - g_{\theta}(f_{\phi}(\tilde{x}))\|^2$$

Common corruptions: additive Gaussian noise, random masking (dropout), salt-and-pepper noise.

Insight: DAEs learn to capture the **manifold structure** of the data—projecting noisy points back onto the data manifold.

Sparse autoencoder

Add an ℓ_1 penalty on activations:

$$\mathcal{L}_{\text{sparse}} = \|x - \hat{x}\|^2 + \lambda \sum_l \|h^{(l)}\|_1$$

This encourages each hidden unit to respond to a specific feature, improving interpretability.

Connection to generative modeling

DAEs are closely related to **score matching** and **denoising score matching**, which underpin diffusion models (covered in Volume 03).

L07: Latent Variable Models and the Probabilistic Perspective

Learning objectives

- Formulate generative modeling with latent variables.
- Introduce variational inference as an approximation strategy.

Latent variable model

Assume data is generated by sampling latent $z \sim p(z)$, then $x \sim p_{\theta}(x | z)$:

$$p_{\theta}(x) = \int p_{\theta}(x | z) p(z) dz$$

Typically $p(z) = \mathcal{N}(0, I)$ and $p_{\theta}(x | z) = \mathcal{N}(\mu_{\theta}(z), \sigma_{\theta}^2 I)$.

Intractability

The marginal likelihood $p_{\theta}(x) = \int p_{\theta}(x | z) p(z) dz$ is generally **intractable** for neural network decoders. We cannot compute $\log p_{\theta}(x)$ exactly.

Variational inference

Introduce an approximate posterior $q_\phi(z | x)$ and optimize a lower bound on $\log p_\theta(x)$ instead of the marginal directly.

Key identity

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\text{KL}}(q_\phi(z | x) \| p_\theta(z | x))$$

Since $D_{\text{KL}} \geq 0$, $\mathcal{L}$ is a **lower bound** on the log-likelihood.

L08: Variational Autoencoders and the Reparameterization Trick

Learning objectives

- Describe the VAE architecture.
- Explain the reparameterization trick for gradient estimation.

VAE architecture

flowchart LR

```
X["Input x"] --> ENC["Encoder q_φ(z|x)"]
ENC --> MU["μ_φ(x)"]
ENC --> SIG["σ_φ(x)"]
MU --> REP["Reparameterize: z = μ + σ·ε"]
SIG --> REP
EPS["ε ~ N(0,I)"] --> REP
REP --> DEC["Decoder p_θ(x|z)"]
DEC --> XHAT["Reconstruction x̂"]
```

The encoder outputs parameters of a Gaussian:

$$q_\phi(z | x) = \mathcal{N}(z; \mu_\phi(x), \sigma_\phi^2(x))$$

Reparameterization trick

To backpropagate through stochastic sampling:

$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

This expresses z as a deterministic function of ϕ and noise ε , enabling gradient flow.

Decoder

The decoder parameterizes $p_\theta(x | z)$. For continuous data:

$$p_\theta(x | z) = \mathcal{N}(x; \mu_\theta(z), \sigma_{\text{dec}}^2 I)$$

Reconstruction loss becomes (up to constants) $\|x - \mu_\theta(z)\|^2$.

L09: Evidence Lower Bound (ELBO) Derivation

Learning objectives

- Derive the ELBO step by step.
- Interpret the reconstruction and KL terms.

Full derivation

Starting from the KL identity:

$$D_{\text{KL}}(q_\phi(z | x) \| p_\theta(z | x)) = \mathbb{E}_{q_\phi} [\log q_\phi(z | x) - \log p_\theta(z | x)]$$

Using Bayes' rule, $\log p_\theta(z | x) = \log p_\theta(x | z) + \log p(z) - \log p_\theta(x)$:

$$\log p_\theta(x) = \mathbb{E}_{q_\phi} [\log p_\theta(x | z)] - D_{\text{KL}}(q_\phi(z | x) \| p(z)) + D_{\text{KL}}(q_\phi(z | x) \| p_\theta(z | x))$$

Dropping the non-negative KL term yields the **ELBO**:

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x | z)] - D_{\text{KL}}(q_\phi(z | x) \| p(z))$$

Gaussian case

With $q_\phi(z | x) = \mathcal{N}(\mu_\phi, \sigma_\phi^2 I)$ and $p(z) = \mathcal{N}(0, I)$:

$$D_{\text{KL}} = \frac{1}{2} \sum_{j=1}^k (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1)$$

Training objective

$$\mathcal{L}_{\text{VAE}} = \frac{1}{N} \sum_{i=1}^N [\|x^{(i)} - \mu_\theta(z^{(i)})\|^2 + \beta \cdot D_{\text{KL}}(q_\phi(z | x^{(i)}) \| p(z))]$$

where $z^{(i)} = \mu_\phi(x^{(i)}) + \sigma_\phi(x^{(i)}) \odot \varepsilon^{(i)}$.

Interpretation

Term	Role
Reconstruction $\mathbb{E}[\log p_{\theta}(x \mid z)]$	Fidelity of generated samples
KL penalty	Regularizes latent space toward prior; enables sampling $z \sim p(z)$

Key takeaways

- The ELBO trades off reconstruction quality against latent regularization.
- VAEs provide a principled probabilistic framework but often produce blurry images compared to GANs (Volume 02).

Volume 01 summary

Lecture	Core concept
L01	Generative vs. discriminative; taxonomy
L02	Neural nets, backprop, optimization
L03	CNNs: LeNet, VGG, ResNet
L04	Autoencoder architecture
L05	Reconstruction loss
L06	Denoising and sparse autoencoders
L07	Latent variable models
L08	VAE + reparameterization trick
L09	ELBO derivation

Next: [Volume 02 — Advanced VAEs and GANs](#)